

Project Report — Trajectory-Controlled Manipulator (SCARA-style) to Draw Accurate Arcs

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Duration target: 7–8 weeks

Funding: Student-funded (baseline budget shown)

Executive Summary

This project proposes the design and implementation of a compact 2-DOF SCARA-style manipulator (rotary shoulder + rotary elbow + optional pen-lift) to execute smooth, time-parameterized trajectories for accurate arc drawing. The manipulator is powered by low-cost NEMA17 steppers, DRV8825 drivers, an Arduino Uno with CNC shield, and a regulated SMPS. Trajectory planning is performed using quintic polynomials, with validation via webcam-based path measurement. The system is designed for affordability, reproducibility, and educational value.

Objectives

1. Implement inverse kinematics for a 2R planar manipulator and trajectory planning (quintic polynomials with time scaling).
 2. Build a physical manipulator capable of drawing arcs of varying radii (20–100 mm and above).
 3. Measure and report RMS path error and repeatability across multiple trials.
 4. Deliver CAD files, firmware, test data, and a demonstration video at the end of Week-8.
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Scope & Constraints

- Workspace limited to 2D planar drawing.
 - Budget constraint: **₹8,500–9,000 (minimum MRP)** for essential components.
 - Time constraint: completion within **7–8 weeks**, including multiple trial/error iterations.
 - Accuracy target: **2–4 mm RMS** (with current hardware).
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System Architecture

- **Mechanical Structure:** Two rotary joints (shoulder θ_1 , elbow θ_2). Frame constructed with 6 mm acrylic sheet and 3D printed link couplers. Smooth rods or bearings used for support. Pen mounted to end effector; spring-based pen-lift optional.
- **Actuation:** 2 $\times$ NEMA17 stepper motors, driven by GT2 belt and pulley transmission.
- **Electronics:** Arduino Uno, CNC Shield V3, 2 $\times$ DRV8825 drivers, 12 V 5 A SMPS. Breadboard and jumper wires for initial wiring.
- **Software:** Trajectory planning in Python/MATLAB; firmware execution via Arduino. Data logged for analysis.
- **Validation:** Webcam and graph paper for RMS error analysis using OpenCV.

Hardware Bill of Materials (BOM)

Prices shown are minimum MRP values (India, September 2025).

Item	Quantity	Unit Price (₹)	Total (₹)
NEMA17 stepper motor (4.2 kg-cm)	2	671	1,342
DRV8825 stepper driver	2	83	166
Arduino Uno R3	1	350	350
CNC Shield V3	1	163	163
GT2 belt & pulley kit	1	1,775	1,775
Smooth rods + LM8UU bearings	1 set	2,000	2,000
12 V 5 A SMPS	1	1,000	1,000
Acrylic sheet (6 mm, 1 sq. ft.)	1	580	580
Breadboard + jumper wires	1	300	300
Wiring, connectors, fasteners	Lump sum	800	800
Total (Essential)			8,476

Optional (if budget allows):

- Webcam for validation = ₹1,500
- Encoder-based drivers = ₹2,000–6,000
- MGN12 linear rails (per axis) = ₹3,000–6,000

Cost Analysis

- Essential components total: **₹8,476.**

- Well within the ₹15,000–20,000 expected budget range for M.Tech first-semester projects.
 - Additional optional upgrades can be considered depending on results.
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Procurement Priorities

1. Stepper motors and drivers.
 2. GT2 belt kit and rods/bearings.
 3. Acrylic frame fabrication and 3D printed parts.
 4. Arduino Uno, CNC Shield, SMPS, and accessories.
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Simulation & Validation Plan

- Simulate inverse kinematics and trajectory planning in MATLAB/Python.
 - Validate torque limits for stepper motors.
 - Perform dry runs at low speed before testing with a pen.
 - Use camera-based measurement to calculate RMS error.
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8-Week Timeline

- **Week 1:** Simulation of kinematics & trajectory, finalize CAD.
 - **Week 2:** Procure parts, begin fabrication of frame.
 - **Week 3:** Assemble motors, rods, bearings, and belts.
 - **Week 4:** Wire electronics, basic motion demo.
 - **Week 5:** Implement trajectory planning in firmware, test slow arcs.
 - **Week 6:** Tune acceleration, jerk, and belt tension; repeatability tests.
 - **Week 7:** Optimize accuracy, integrate webcam measurement.
 - **Week 8:** Final validation, error analysis, documentation, and demonstration.
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Task Distribution

- **Bharathwaj Manoharan:** Trajectory algorithms, simulation, firmware integration, data analysis.
 - **Vandhiya Devan:** CAD design, fabrication, 3D printing, mechanical assembly.
 - **Mohamed Rizvan:** Electronics wiring, motor tuning, safety, and test protocol execution.
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Risks & Mitigation

- **Stepper skipping at high acceleration:** Mitigate with conservative trajectory parameters and microstepping.
- **Frame flex or belt slack:** Use rigid mounting and belt tensioners.

- **Power instability:** Use regulated SMPS and fuses.
 - **Measurement error:** Calibrate camera with grid paper and ensure stable lighting.
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Deliverables

1. CAD files and STL models.
 2. Arduino firmware and trajectory scripts.
 3. Test logs with RMS error analysis.
 4. Final video demonstration and slides.
 5. Procurement sheet with vendor details.
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Approval Request

We request faculty approval to proceed with the proposed SCARA-style manipulator project. Funding will be student-backed (~₹8,476 baseline). Faculty support is requested for lab access and mid-point review (Week-4).